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function MaxProductOfThree(array)

    if length of array < 3

        return "Error: Array must have at least 3 elements"

    return FindExtremes(array, 0, -infinity, -infinity, -infinity,
infinity, infinity)

end function

function FindExtremes(array, index, max1, max2, max3, min1, min2)

    if index == length of array

        return max(max1 * max2 * max3, min1 * min2 * max1)

    num = array[index]

    if num > max1
        max3 = max2
        max2 = max1
        max1 = num
    else if num > max2
        max3 = max2
        max2 = num
    else if num > max3
        max3 = num

    if num < min1
        min2 = min1
        min1 = num
    else if num < min2
        min2 = num

    return FindExtremes(array, index + 1, max1, max2, max3, min1, min2)

end function

```